

COMPUTER NETWORK

Lecture 6

1

Transport Layer

- It is responsible for the actual mechanics for:
 - Session establishment
 - Session termination
 - Session management and control by:
 - Segmentation
 - Dividing data into small part called segments
 - Sequencing
 - Give a serial number to each segment

Transport Layer

❑ Error detection

Take every segment and apply on it CRC (Cyclic Redundancy Check) equation to give CRC result (4 byte), which send with segment, the receiver when receive that segment and apply on CRC on it and compare the result with the receiving CRC result if the two are not same then the data has error and drop it.

Transport Layer

- Error correction

When the segment arrive to the receiver with error, the receiver drop it and order the sender to resend the segment again

- Reliability

Making sure that the segment are correctly received

- Flow control

Transport Layer

□ It is responsible for :

- Segmentation
- Error detection.
- Session address

Application HTTP FTP RTP
Transportation TCP UDP
Internet IPv4 IPv6
Network Access WiFi ATM Ethernet Frame Relay

Transport Layer

- Some application work with TCP and other work with UDP

TCP	UDP
HTTP HTTPS SMTP POP3 (retrieve mail) FTP Telnet SSH (secure shell) DNS (Server ----- root DNS)	RTP (video , voice , games) DHCP TFTP SNMP DNS (user----- DNS server)

IP Network Components

○ Some Registered Ports (netstat)

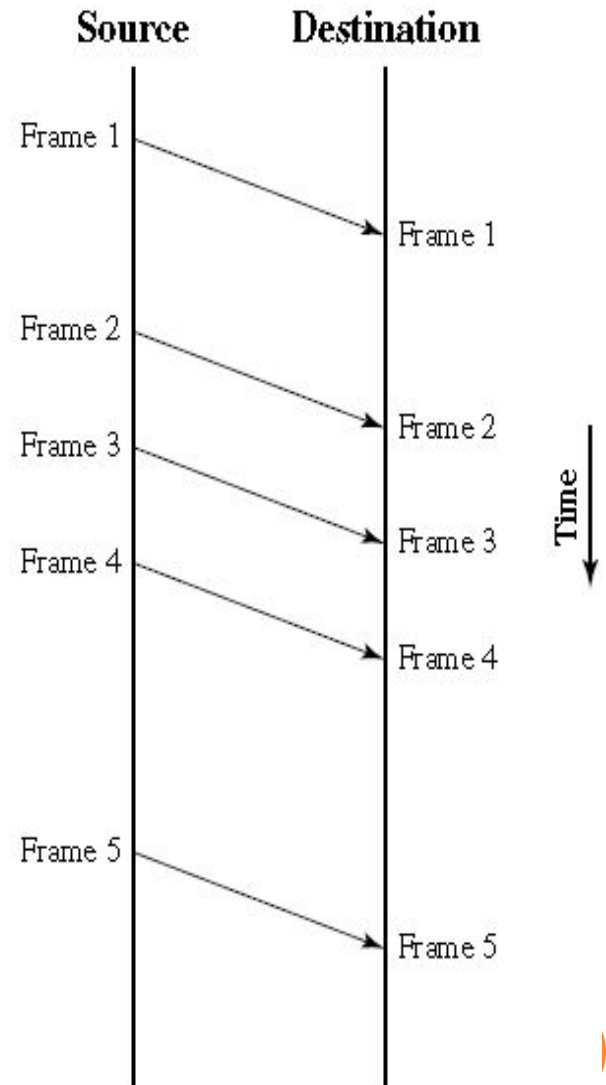
Port no	Application
20,21	FTP
22	SSH
23	Telnet
25	SMTP
80	HTTP
110	POP3
53	DNS
443	HTTPS

Flow Control

- Flow control is a technique for assuring that a transmitting entity does not overwhelm a receiving entity with data
- The receiving entity typically allocates a data buffer of some maximum length for a transfer
- When data are received, the receiver must do a certain amount of processing before passing the data to the higher-level software. In the absence of flow control, the receiver's buffer may fill up and overflow while it is processing old data.

Flow Control

- To begin, we examine mechanisms for flow control in the absence of errors. The model we will use is depicted in the Figure
- Each arrow represents a single frame transiting a data link between two stations
- The data are sent in a sequence of frames, with each frame containing a portion of the data and some control information



Flow Control

- The time it takes for a station to emit all of the bits of a frame onto the medium is the **transmission time**
- The **propagation time** is the time it takes for a bit to traverse the link between source and destination.
- The data are sent in a sequence of frames, with each frame containing a portion of the data and some control information
- Firstly, we assume that all frames that are transmitted are successfully received; no frames are lost and none arrive with errors. Furthermore, frames arrive in the same order in which they are sent.

Stop-and-Wait Flow Control

- The simplest form of flow control, known as stop-and-wait flow control, works as follows:
 - A source entity transmits a frame.
 - After the destination entity receives the frame, it indicates its willingness to accept another frame by sending back an acknowledgment to the frame just received.
 - The source must wait until it receives the acknowledgment before sending the next frame.
 - The destination can thus stop the flow of data simply by withholding acknowledgment

Sliding-Window Flow Control

- The essence of the problem described so far is that only one frame at a time can be in transit.
- In situations where the bit length of the link is greater than the frame length serious inefficiencies result. Efficiency can be greatly improved by allowing multiple frames to be in transit at the same time
- Let us work for two stations, A and B, connected via a full-duplex link

Sliding-Window Flow Control

- Station B allocates buffer space for W frames. Thus, B can accept W frames, and A is allowed to send W frames without waiting for any acknowledgments.
- To keep track of which frames have been acknowledged, each is labeled with a sequence number. B acknowledges a frame by sending an acknowledgment that includes the sequence number of the next frame expected.
- This acknowledgment also implicitly announces that B is prepared to receive the next W frames, beginning with the number specified.

Sliding-Window Flow Control

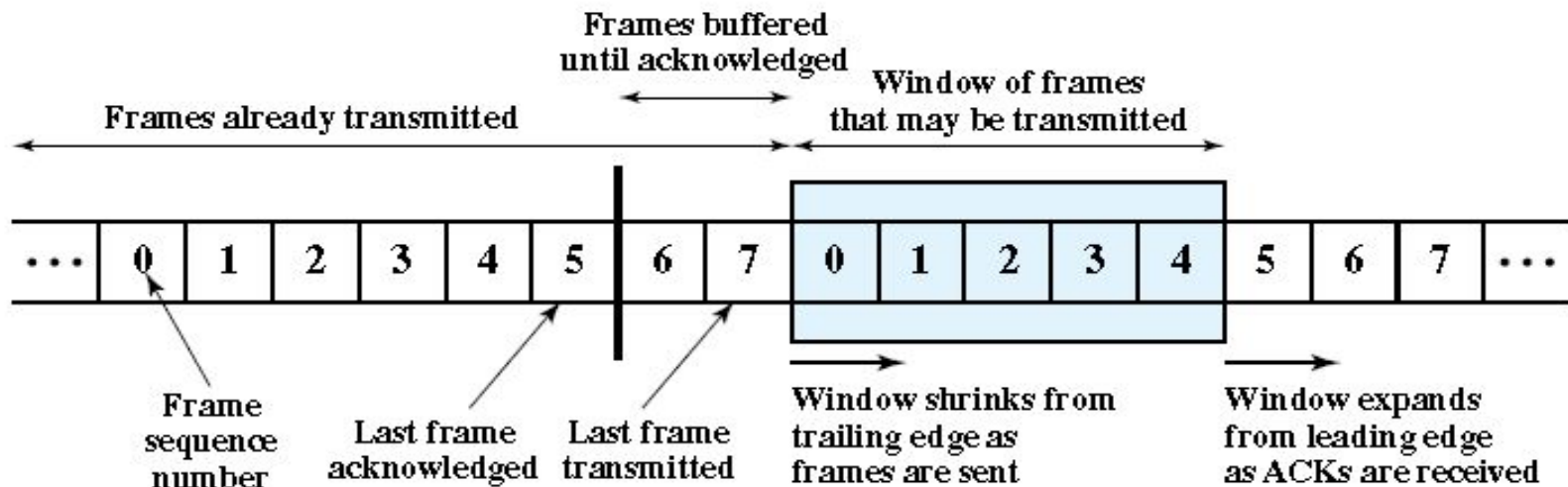
- This scheme can also be used to acknowledge multiple frames.
For example, B could receive frames 2, 3, and 4 but withhold acknowledgment until frame 4 has arrived. By then returning an acknowledgment with sequence number 5, B acknowledges frames 2, 3, and 4 at one time.
 - A maintains a list of sequence numbers that it is allowed to send, and B maintains a list of sequence numbers that it is prepared to receive. Each of these lists can be thought of as a *window of frames*. The operation is referred to as ***sliding-window flow control***.
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Sliding-Window Flow Control

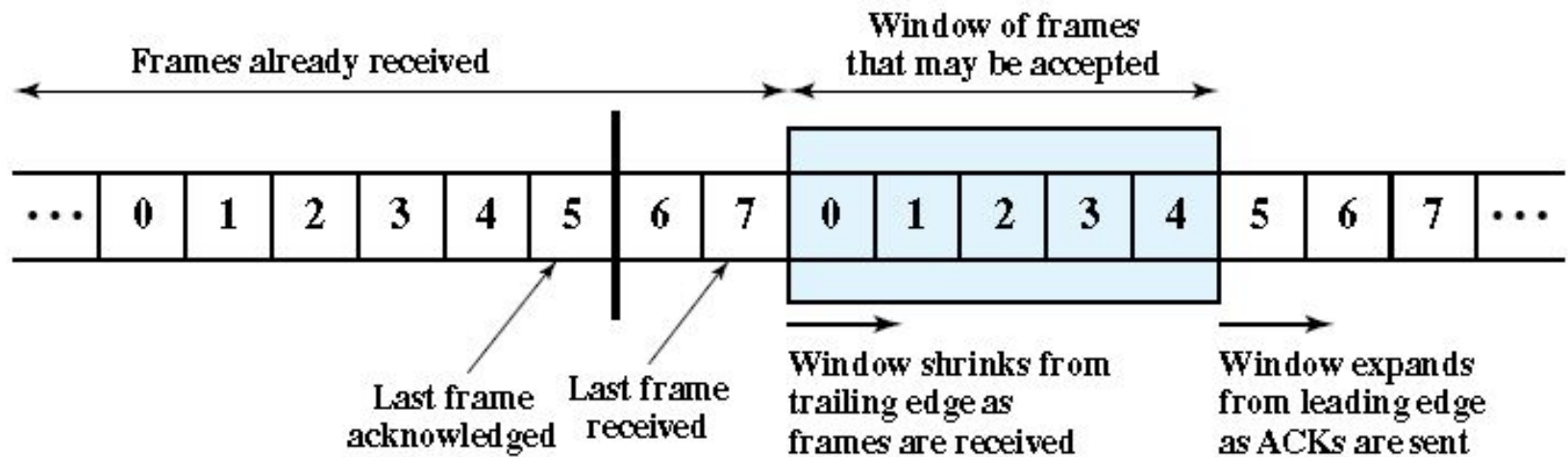
- Because the sequence number to be used occupies a field in the frame, it is limited to a range of values. For example, for a 3-bit field, the sequence number can range from 0 to 7. Accordingly, frames are numbered modulo 8; that is, after sequence number 7, the next number is 0.
- The following Figure is a useful way of depicting the sliding-window process. It assumes the use of a 3-bit sequence number, so that frames are numbered sequentially from 0 through 7, and then the same numbers are reused for subsequent frames



Sliding-Window Depiction



(a) Sender's perspective



(b) Receiver's perspective

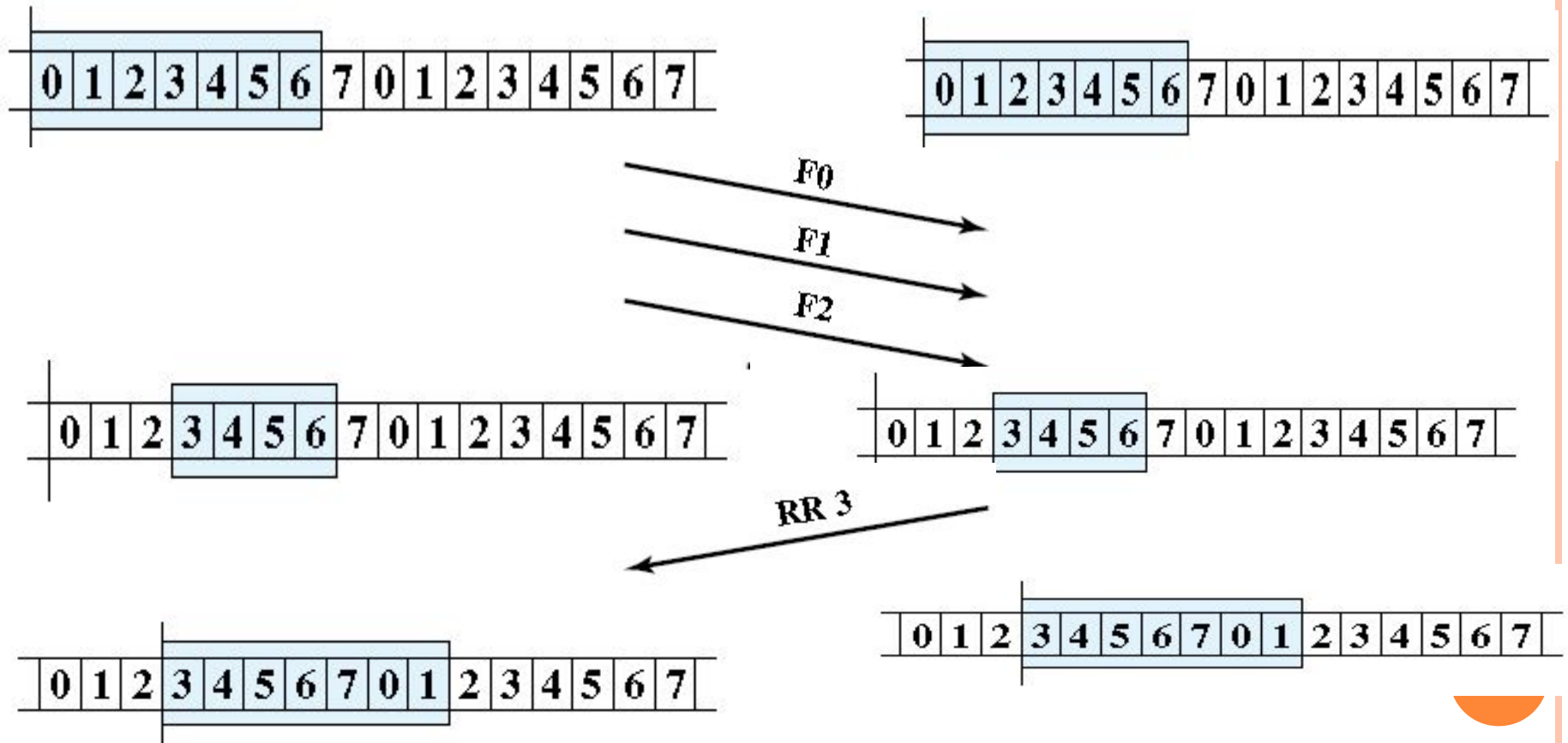
Sliding-Window Depiction

- The shaded rectangle indicates the frames that may be sent; in this figure, the sender may transmit five frames, beginning with frame 0.
- Each time a frame is sent, the shaded window shrinks; each time an acknowledgment is received, the shaded window grows.
- Frames between the vertical bar and the shaded window have been sent but not yet acknowledged. As we shall see, the sender must buffer these frames in case they need to be retransmitted.

Example of a Sliding-Window Protocol

Source system A

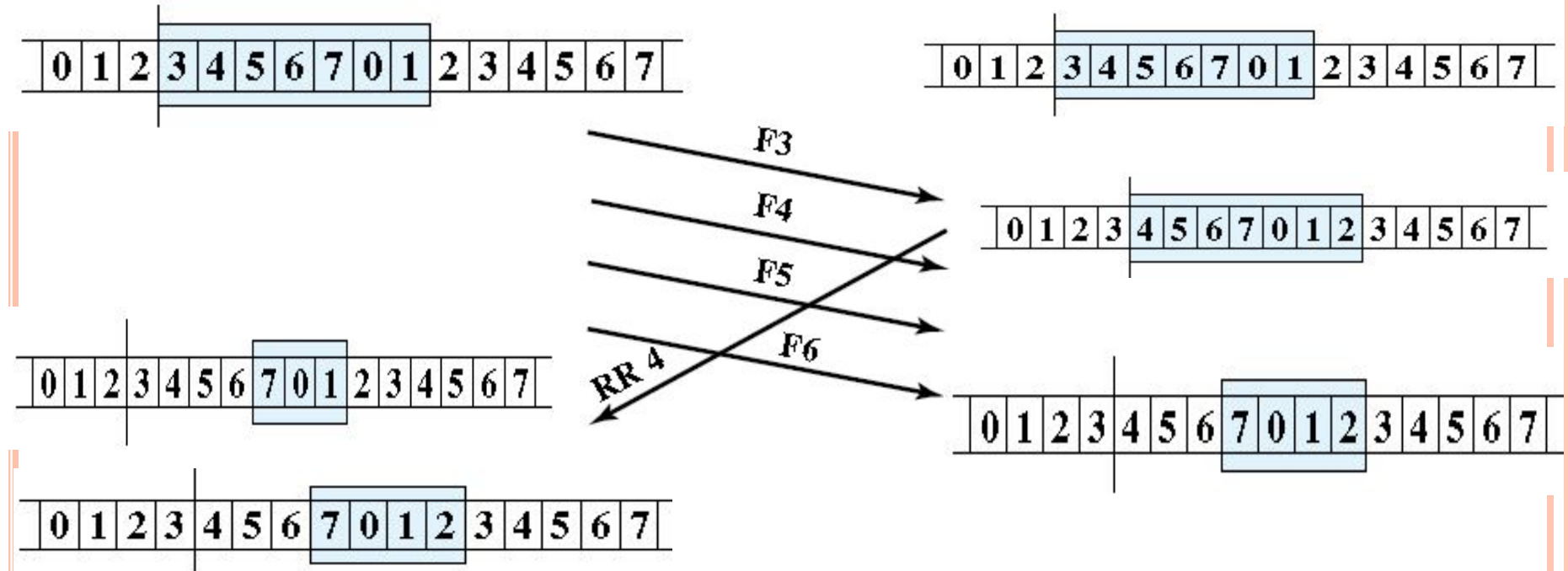
Destination system B



Example of a Sliding-Window Protocol

Source system A

Destination system B



Sliding-Window Control Protocol

- Most data link control protocols also allow a station to cut off the flow of frames from the other side by sending a Receive Not Ready (RNR) message, which acknowledges former frames but forbids transfer of future frames.
- Thus, RNR 5 means “I have received all frames up through number 4 but am unable to accept any more at this time.”
- we have discussed transmission in one direction only.



Sliding-Window Control Protocol

- If two stations exchange data, each needs to maintain two windows, one for transmit and one for receive, and each side needs to send the data and acknowledgments to the other.
- To provide efficient support for this requirement, a feature known as **piggybacking** is typically provided. Each data frame includes a field that holds the sequence number of that frame plus a field that holds the sequence number used for acknowledgment.



Sliding-Window Control Protocol

- if a station has data to send and an acknowledgment to send, it sends both together in one frame, saving communication capacity
- if a station has an acknowledgment but no data to send, it sends a separate acknowledgment frame, such as RR or RNR.
- If a station has data to send but no new acknowledgment to send, it must repeat the last acknowledgment sequence number that it sent.
- This is because the data frame includes a field for the acknowledgment number, and some value must be put into that field. When a station receives a duplicate acknowledgment, it simply ignores it.

Problem

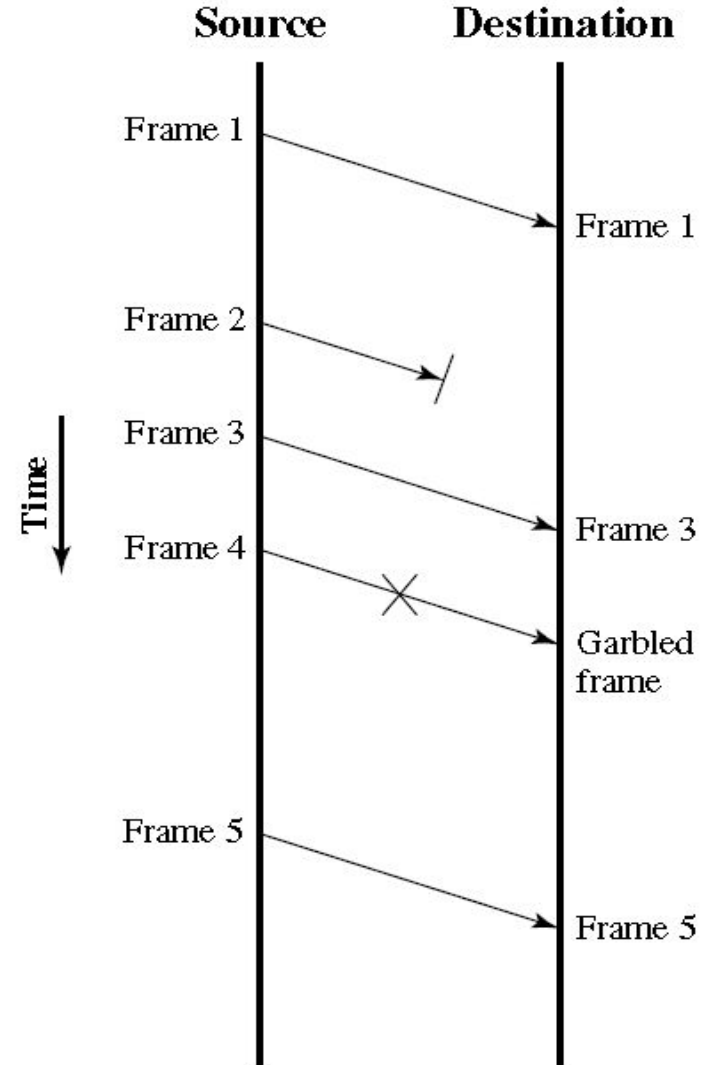
Two neighboring nodes (A and B) use a sliding-window flow control protocol. The sliding window mechanism with a window size of 4 is used. Assuming A is transmitting and B is receiving, show the window positions at both A and B for the following succession of events:

- a) Before A sends any frames.
- b) After A sends frames 0, 1, 2 and B acknowledges 0, 1 and the ACK is received by A.
- c) After A sends frames 3, 4, 5 and B acknowledges 4 and the ACK is received by A.



ERROR CONTROL

- Error control refers to mechanisms to detect and correct errors that occur in the transmission of frames.
- **Lost frame:** A frame fails to arrive at the other side.
- **Damaged frame:** A recognizable frame does arrive, but some of the bits are in Error.



(b) Transmission with losses and errors

ERROR CONTROL

- Error control refers to mechanisms to detect and correct errors that occur in the transmission of frames.
- The most common technique for error control is **automatic repeat request (ARQ)**; the effect of ARQ is to turn an unreliable data link into a reliable one.
- Three versions of ARQ have been standardized:
 - Stop-and-wait ARQ
 - Go-back-N ARQ
 - Selective-reject ARQ



Stop-and-Wait ARQ

- Stop-and-wait ARQ is based on the stop-and-wait flow control technique outlined previously
- The source station transmits a single frame and then must await an acknowledgment (ACK). No other data frames can be sent until the destination station's reply arrives at the source station.
- Two sorts of errors could occur. First, the frame that arrives at the destination could be damaged. The receiver detects this by using the error-detection technique.

Stop-and-Wait ARQ

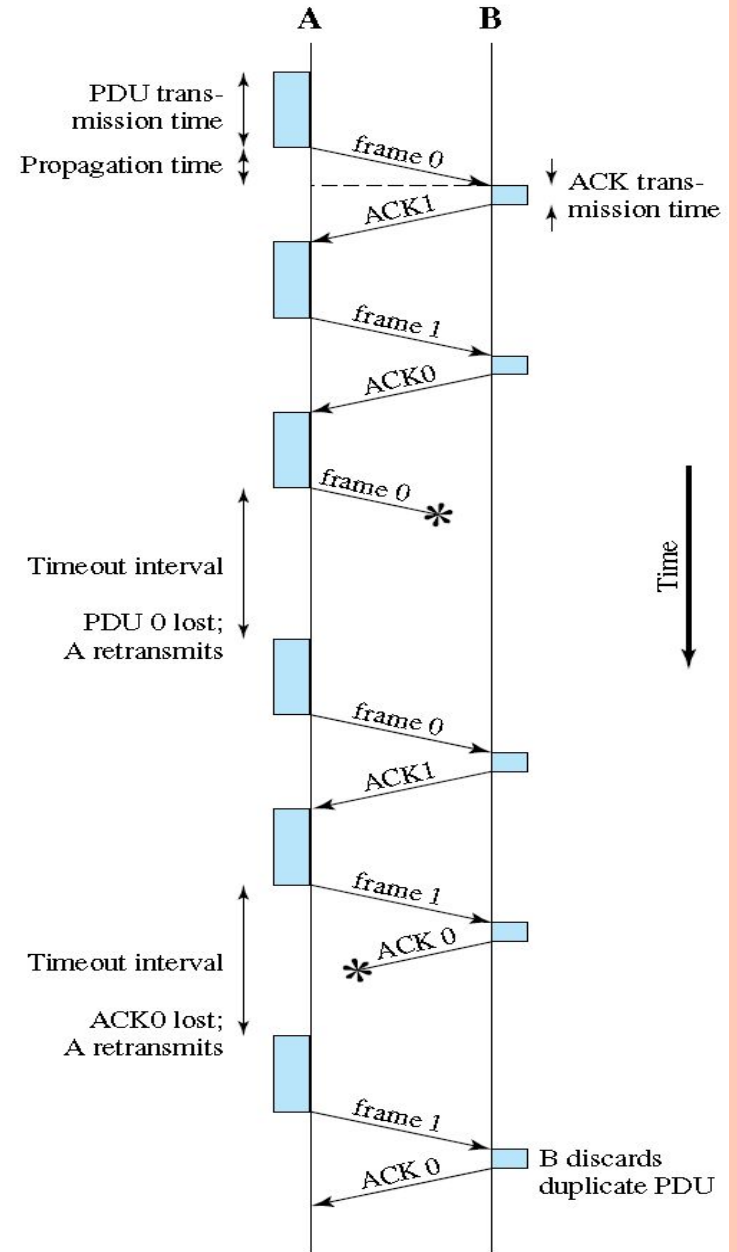
- To account for this possibility, the source station is equipped with a timer. After a frame is transmitted, the source station waits for an acknowledgment.
- If no acknowledgment is received by the time that the timer expires, then the same frame is sent again. Note that this method requires that the transmitter maintain a copy of a transmitted frame until an acknowledgment is received for that frame.

Stop-and-Wait ARQ

- The second sort of error is a damaged acknowledgment. Consider the following situation. Station A sends a frame. The frame is received correctly by station B, which responds with an acknowledgment (ACK).
- The ACK is damaged in transit and is not recognizable by A, which will therefore time out and resend the same frame. This duplicate frame arrives and is accepted by B.
- B has therefore accepted two copies of the same frame as if they were separate

Stop-and-Wait ARQ

- To avoid this problem, frames are alternately labeled with 0 or 1, and positive acknowledgments are of the form ACK0 and ACK1. In keeping with the sliding-window convention, an ACK0 acknowledges receipt of a frame numbered 1 and indicates that the receiver is ready for a frame numbered 0



Go-Back-N ARQ

- The form of error control based on sliding-window flow control that is most commonly used is called go-back-N ARQ.
- In this method, a station may send a series of frames sequentially numbered modulo some maximum value. The number of unacknowledged frames outstanding is determined by window size, using the sliding-window flow control technique.
- While no errors occur, the destination will acknowledge incoming frames as usual (RR = receive ready, or piggybacked acknowledgment).

Go-Back-N ARQ

- ❑ If the destination station detects an error in a frame, it may send a negative acknowledgment (REJ = reject) for that frame, as explained in the following rules.
- ❑ The destination station will discard that frame and all future incoming frames until the frame in error is correctly received
- ❑ Thus, the source station, when it receives a REJ, must retransmit the frame in error plus all succeeding frames that were transmitted in the interim.

Go-Back-N ARQ

- Suppose that station A is sending frames to station B. After each transmission, A sets an acknowledgment timer for the frame just transmitted. Suppose that B has previously successfully received frame $(i-1)$ and A has just transmitted frame i . The go-back-N technique takes into account the following contingencies:
- **Damaged frame**. If the received frame is invalid B discards the frame and takes no further action as the result of that frame. There are two subcases:

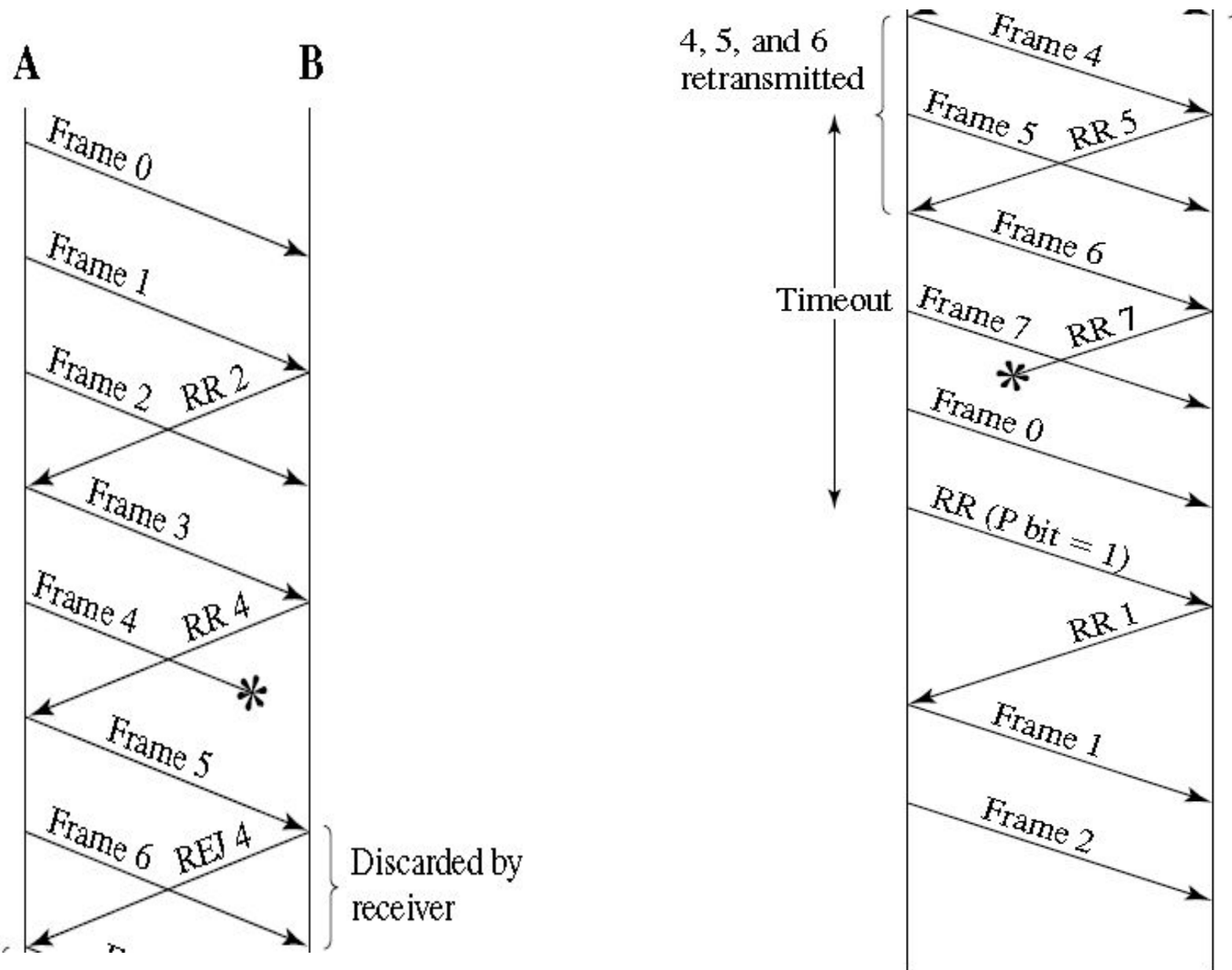
Go-Back-N ARQ

1. Within a reasonable period of time, A subsequently sends frame (i+1). B receives frame (i+1) out of order and sends a REJ i. A must retransmit frame i and all subsequent frames.
2. A does not soon send additional frames. B receives nothing and returns neither an RR nor a REJ. When A's timer expires, it transmits an RR frame that includes a bit known as the P bit, which is set to 1. B interprets the RR frame with a P bit of 1 as a command that must be acknowledged by sending an RR indicating the next frame that it expects, which is frame i. When A receives the RR, it retransmits frame i. Alternatively, A could just retransmit frame i when its timer expires.

Go-Back-N ARQ

- Damaged RR. There are two subcases:
 - B receives frame i and sends RR which suffers an error in transit. Because acknowledgments are cumulative, it may be that A will receive a subsequent RR to a subsequent frame and that it will arrive before the timer associated with frame i expires.
 - If A's timer expires, it transmits an RR command. It sets another timer, called the P-bit timer. If B fails to respond to the RR command, or if its response suffers an error in transit, then A's P-bit timer will expire. At this point, A will try again by issuing a new RR command and restarting the P-bit timer.
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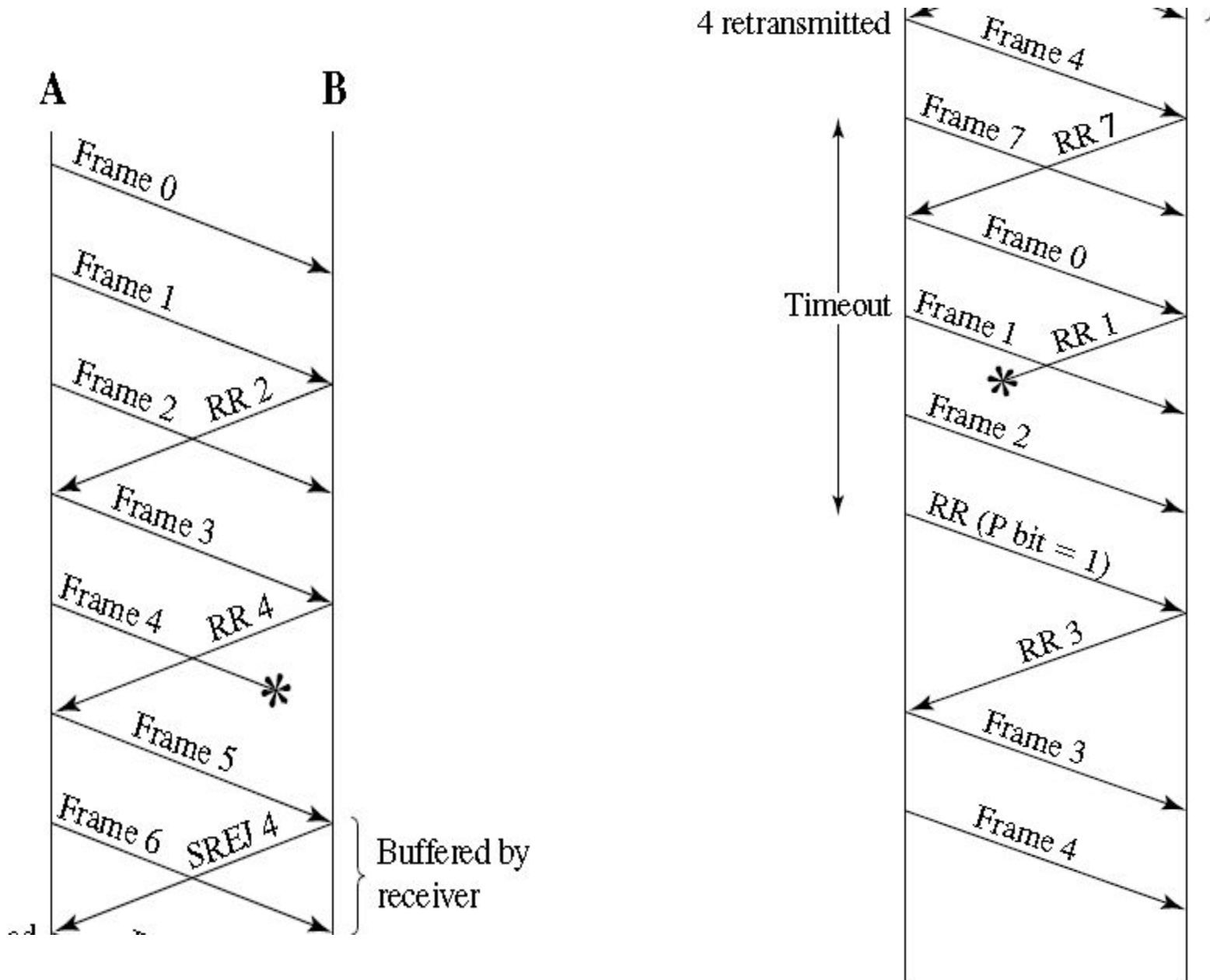
Go-Back-N ARQ



Selective-Reject ARQ

- With selective-reject ARQ, the only frames retransmitted are those that receive a negative acknowledgment, in this case called SREJ, or those that time out.
- Selective reject would appear to be more efficient than go-back-N, because it minimizes the amount of retransmission.
- On the other hand, the receiver must maintain a buffer large enough to save post-SREJ frames until the frame in error is retransmitted and must contain logic for reinserting that frame in the proper sequence. Because of such complications, select-reject ARQ is much less widely used than go-back-N ARQ.

Selective-Reject ARQ



Problem

- 1) Two neighbouring nodes (A and B) use a sliding-window protocol with a 3-bit sequence number. As the ARQ mechanism, go-back-N is used with a window size of 4. Assuming A is transmitting and B is receiving, show the window positions for the following succession of events:
 - a) Before A sends any frames
 - b) After A sends frames 0, 1, 2 and receives acknowledgement from B for frames 0 and 1
 - c) After A sends frames 3, 4, 5 and B acknowledges 4 and the ACK is received by A.



Problem

- 1) Draw the sender and receiver windows for a system using selective reject ARQ given the following:
 - a) Frame 0 is sent; frame 0 is acknowledged
 - b) Frames 1 and 2 are sent; frames 1 and 2 are acknowledged
 - c) Frames 3, 4 and 5 are sent; NACK 4 is received
 - d) Frames 4, 5, 6 and 7 are sent; frames 4 through 7 are acknowledged.
- 2) Suppose that a selective-reject ARQ is used where $W=4$, Show by example that a 3-bit sequence number is needed

